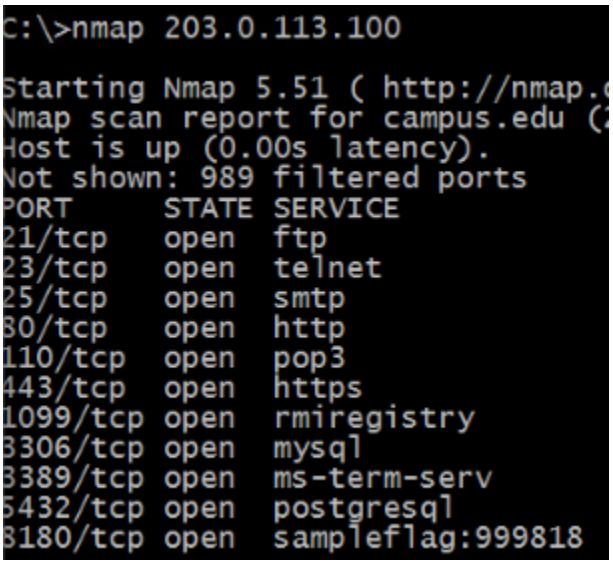
Lab 9.1: Crafting and Deploying Malware Using a Remote Access Trojan (RAT)

Lab 9.1: Part 1 – Launching an Attack

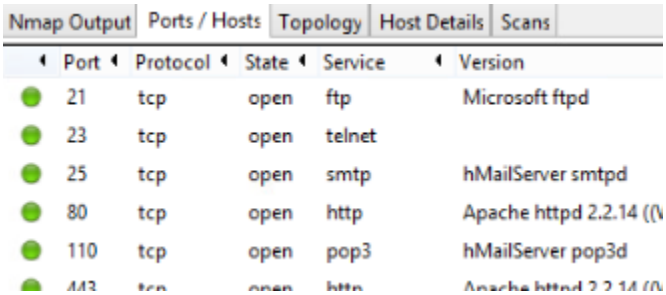
1. Question 2

2. 

```
C:\>nmap 203.0.113.100

Starting Nmap 5.51 ( http://nmap.
Nmap scan report for campus.edu (
Host is up (0.00s latency).
Not shown: 989 filtered ports
PORT     STATE SERVICE
21/tcp   open  ftp
23/tcp   open  telnet
25/tcp   open  smtp
80/tcp   open  http
110/tcp  open  pop3
443/tcp  open  https
1099/tcp open  rmiregistry
3306/tcp open  mysql
3389/tcp open  ms-term-serv
5432/tcp open  postgresql
8180/tcp open  sampleflag:999818
```

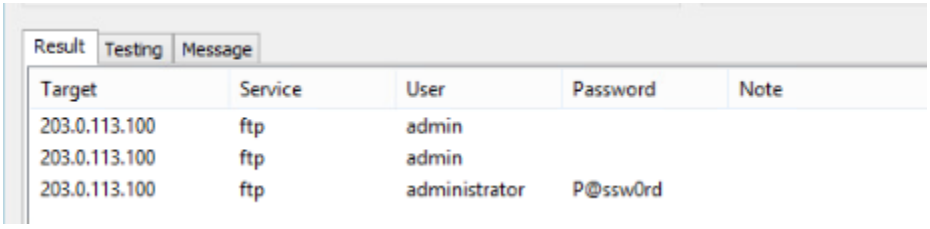
3. Question 4

4. 

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

Port	Protocol	State	Service	Version
21	tcp	open	ftp	Microsoft ftpd
23	tcp	open	telnet	
25	tcp	open	smtp	hMailServer smtpd
80	tcp	open	http	Apache httpd 2.2.14 ((\
110	tcp	open	pop3	hMailServer pop3d
443	tcp	open	http	Apache httpd 2.2.14 ((\

5. Question 11

6. 

Result | Testing | Message

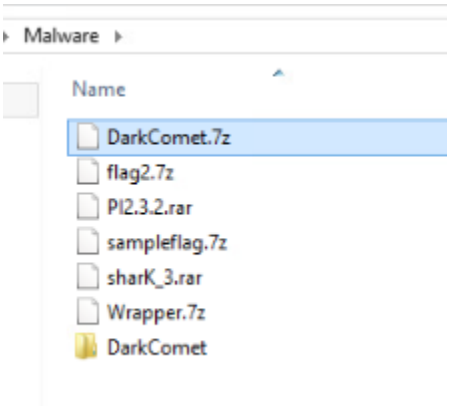
Target	Service	User	Password	Note
203.0.113.100	ftp	admin		
203.0.113.100	ftp	admin		
203.0.113.100	ftp	administrator	P@ssw0rd	

7. What port is being targeted in the attack?

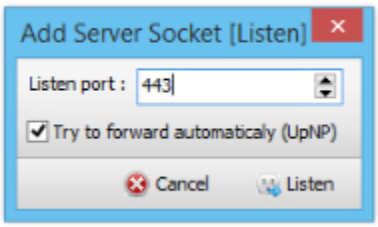
Port 21

Lab 9.1: Part 2 – Building and Deploying the Malware

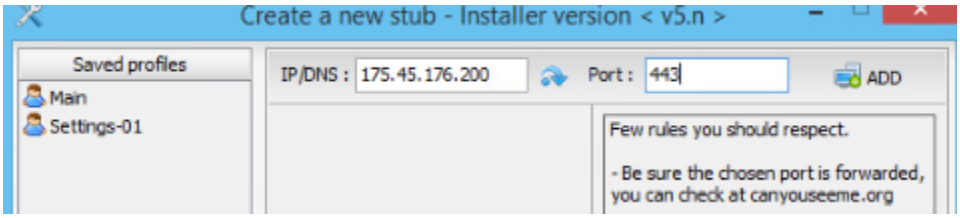
1. Question 8

2. 

3. Question 16

4. 

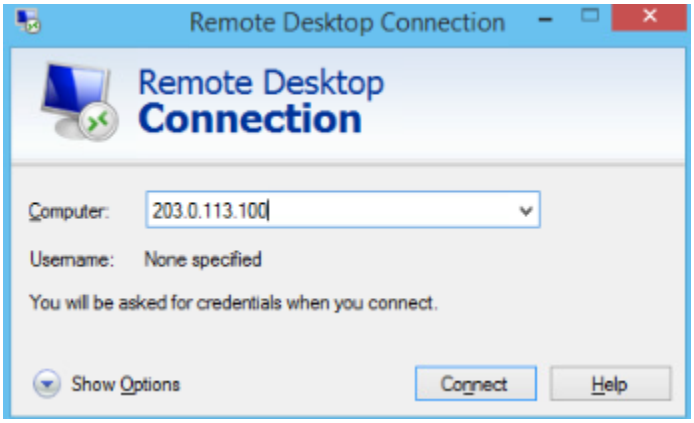
5. Question 18

6. 

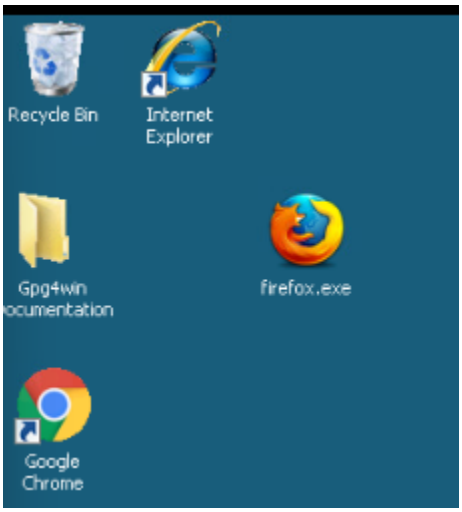
7. Question 25

8.

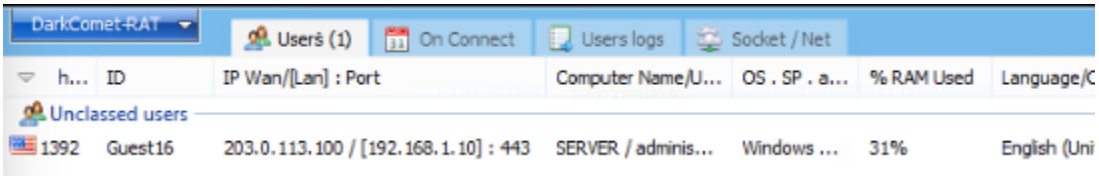
9. Question 27

10. 

11. Question 31

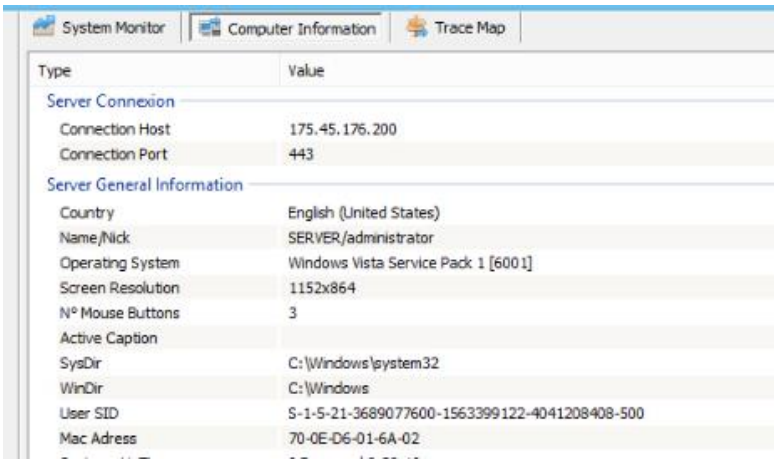
12. 

13. Question 34

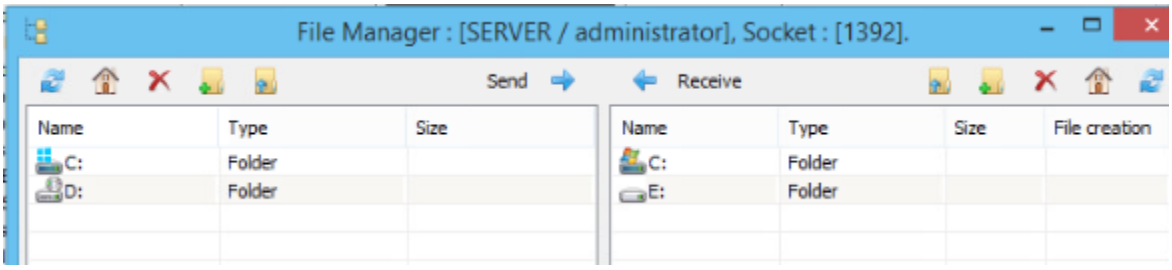


Lab 9.1: Part 3 – Exploiting the Victim

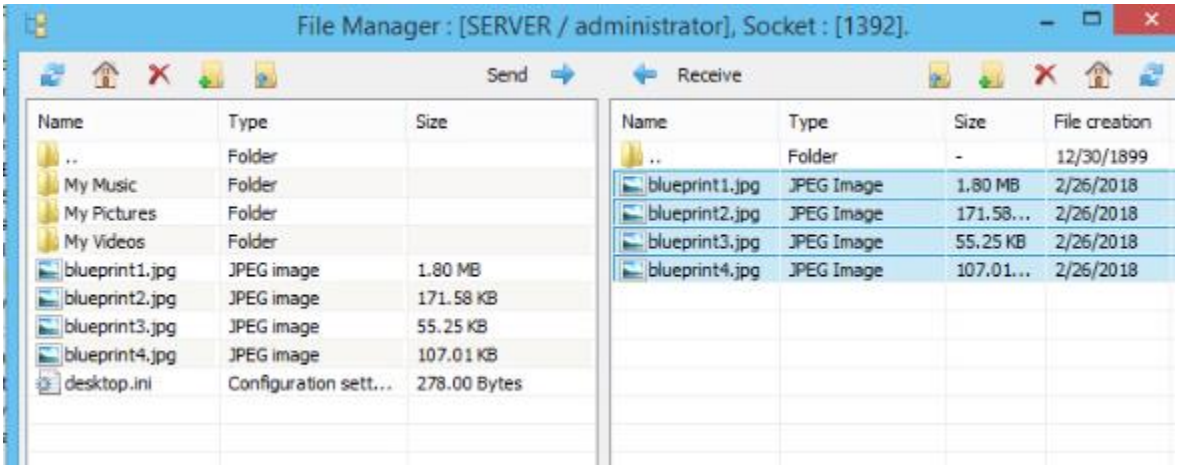
1. Question 3

2. 

3. Question 4

4. 

5. Question 13



Lab 9.1: Part 4 – Lessons Learned

Lessons Learned

Learned

- In this lab, we were able to gain the admin password of the firewall through a brute force attack using a dictionary. Then we created malware using the application DarkComet by hiding it as a Firefox application. We took a copy of the infected application and put it on the desktop of the Windows server to then gain access to the Windows files. I did not know that we could connect to another desktop using

the Microsoft Terminal Service Client remotely. Plus, with the use of brute force I feel like connecting to anyone's computer would be fairly easy.

Surprises & Challenges

- I ran into a problem when trying to crack the password with the brute force application. I had to run it a few times before it gave me the correct output. It kept saying the test was completed yet nothing showed at the bottom, so I messed with the username and exited the app a couple of times to get it to work.
- How do we know the IP address of the firewall?
- When we cracked the password through port 21 since that port was a Microsoft service, is that why we were able to gain access to the Windows server once we logged in remotely?

Feedback

- This lab took me about an hour to complete
- My favorite part of the lab was when we used the brute force technique to gain the password for the Windows server and then access it with that password